

SAMYAK MOON

Bengaluru, Karnataka

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PROFESSIONAL SUMMARY

Backend Software Engineer with 2.8+ years of experience designing and building scalable, secure fintech systems using **Java** and **Spring Boot**. Experienced in **microservices**, **distributed systems**, and **payment/card platforms** with a strong focus on performance, reliability, and system design.

EXPERIENCE

Software Engineer, PayU Digital Labs

April 2025 - Present

Bengaluru, Karnataka

- Designed and owned the **Card Order Workflow** end-to-end using a **maker-checker approval pattern** and **Quartz-based async job scheduling** for Visa and RuPay card generation across multiple clients; optimised for **parallel processing**, reducing card generation time for 10K+ cards by **85%**. Recieved **Extra Mile Award** as Individual Contribution for Card Order Flow.
- Designed **SecureAuthPro**, a **multi-tenant Authentication Authorization** service using **Spring Boot**, **Redis**, and **RSA-256 JWT**; supported 10+ enterprise clients with strict **tenant isolation**, **OTP-based login**, and **Role-Based Access Control** (RBAC) aligned with regulatory security standards.
- Integrated **Atalla Payment HSM** into core services for **PIN generation/verification**, **key management**, and **PCI-DSS compliant** cryptographic operations; achieved **60%** faster transaction processing compared to a general-purpose HSM.
- Built **Wallet-to-Account Transfer (W2A) API** with **idempotency checks**, **concurrency control**, and **retry mechanisms**; implemented a **Quartz-based refund job** to handle failed transactions, ensuring data consistency and **financial reconciliation**.
- Enhanced **Authorization Settlement** flows for **Visa** and **RuPay** network transactions across **e-commerce**, **POS**, and **ATM channels**; reduced processing latency and integrated an **External authentication mechanism** enabling clients to Approve/Reject network transactions in real time.

Associate Software Engineer, PayU Digital Labs

July 2023 - April 2025

Bengaluru, Karnataka

- Implemented **Two-Factor Authentication** (MPIN + OTP) for financial transactions using **Redis TTL-based OTP management**; designed flow to align with **RBI** and **PCI-DSS security expectations**, strengthening **fraud prevention** and access control with low-latency processing.
- Built **Card Adjustment** and **Card Replacement features** using maker-checker workflows with **idempotency** and strict validation logic; enforced one active card per product (Visa/RuPay) per customer, improving operational efficiency by **70%** and reducing customer turnaround time.
- Designed **Customer Segmentation** feature to group customers into dynamic segments and apply **tier-based transactional fee rules**, improving **revenue** configuration flexibility and reducing manual intervention.
- Led **security hardening initiatives** resolving critical vulnerabilities including **CSP**, **CSRF**, **CORS**, **HSTS**, **TLS**, and source-code security issues; reduced audit-reported vulnerabilities by **80%** in collaboration with InfoSec and DevOps teams.

TECHNICAL SKILLS

Programming	Java, Python, SQL
Backend & Distributed Systems	Spring Boot, REST APIs, Microservices, Multi-Tenancy
Databases & Storage	MySQL, PostgreSQL, Redis, Elasticsearch
Cloud & DevOps	AWS, Jenkins, CI/CD
Messaging & Streaming	Kafka, RabbitMQ
Observability & Monitoring	ELK Stack(Elasticsearch, Logstash, Kibana), Grafana
Technologies	Cursor AI (Advanced), LLM-powered coding, Generative AI tools.

EDUCATION

M.Tech in Computer Science

National Institute of Technology Surathkal, Karnataka

2021 - 2023

B.Tech in Computer Science

Government College of Engineering Amravati

2016 - 2020